

Kinematic and Dynamic Modeling of a 4-DOF SCARA Robot

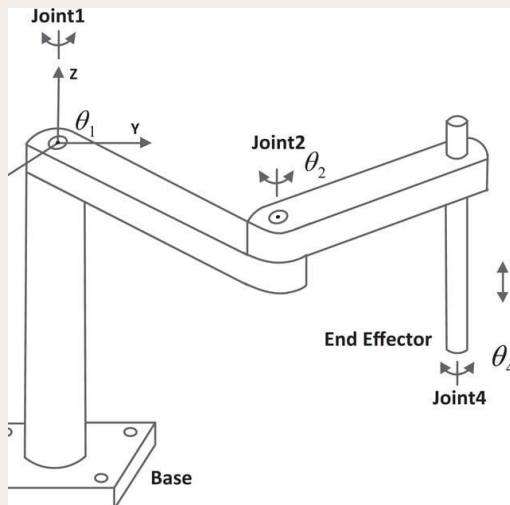
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Shibaura Machine TH550A [1]

SCARA Robot

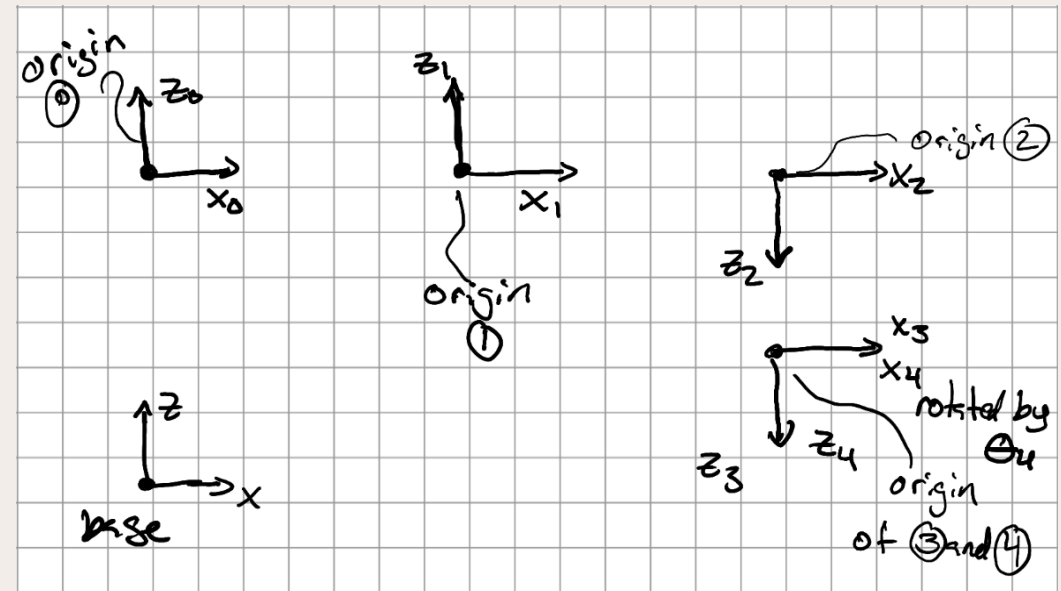
- SCARA stands for Selective Compliance Assembly Robot ARM
- Designed in 1978 and commercialized in 1980 in Japan
- SCARA's selective compliance makes it very effective for pick and place



Typical 4-joint SCARA structure [2]

Mechanical Structure

- 4 Degrees of Freedom (DOF): 3 Revolute (R) and 1 Prismatic (P) joint.
- Configuration: RRP:R (Joints 1, 2, 4 are revolute; Joint 3 is prismatic).
- Coordinate frames assigned using Denavit-Hartenberg (DH) convention.



Elementary Transformations

$$\text{Rot}_z(\theta_i) = \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & 0 \\ \sin\theta_i & \cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{Trans}_z(d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Trans}_x(a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{Rot}_x(\alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_i & -\sin\alpha_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{i-1}^i = \text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i)$$

$$T_{i-1}^i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The Transformation Chain
 - The relative pose between consecutive coordinate frames is defined by a sequence of four basic transformations.
 - Each matrix represents a single degree of freedom or a fixed geometric offset.
- Rotational Components
 - $\text{Rot}_z(\theta_i)$: Rotation about the z_i axis by the joint angle θ_i .
 - $\text{Rot}_x(\theta_i)$: Rotation about the x_i axis by the joint angle θ_i .
- Translational Components
 - $\text{Trans}_z(d_i)$: Translation along the z_i axis by the link offset d_i .
 - $\text{Trans}_x(a_i)$: Translation along the x_i axis by the link offset a_i .

Kinematic Modeling: DH Parameters

Joint i	a_i (m)	α_i (rad)	d_i (m)	θ_i (rad)
1	a_1	0	d_1 (const.)	θ_1 (var.)
2	a_2	π	0	θ_2 (var.)
3	0	0	d_3 (var.)	0
4	0	0	d_4 (const.)	θ_4 (var.)

DH table of a typical SCARA robot

- The DH convention defines the geometry of a robot
- Assigns reference frames to a kinematic chain using a table
- It defines a standard for engineers to follow
- In this case, the $180^\circ(\pi)$ twist at Joint 2 is critical for the downward orientation of the subsequent z-axes.

Forward Kinematics

- Transformation matrix T_0^4 derived by multiplying individual link matrices A_i .
- Describes the end-effector position (x,y,z) and orientation (ϕ) based on joint variables.
- Validation: MATLAB simulation shows the expected butterfly or curved trajectory for combined joint movements.

$$T_0^4 = T_0^1 T_1^2 T_2^3 T_3^4$$

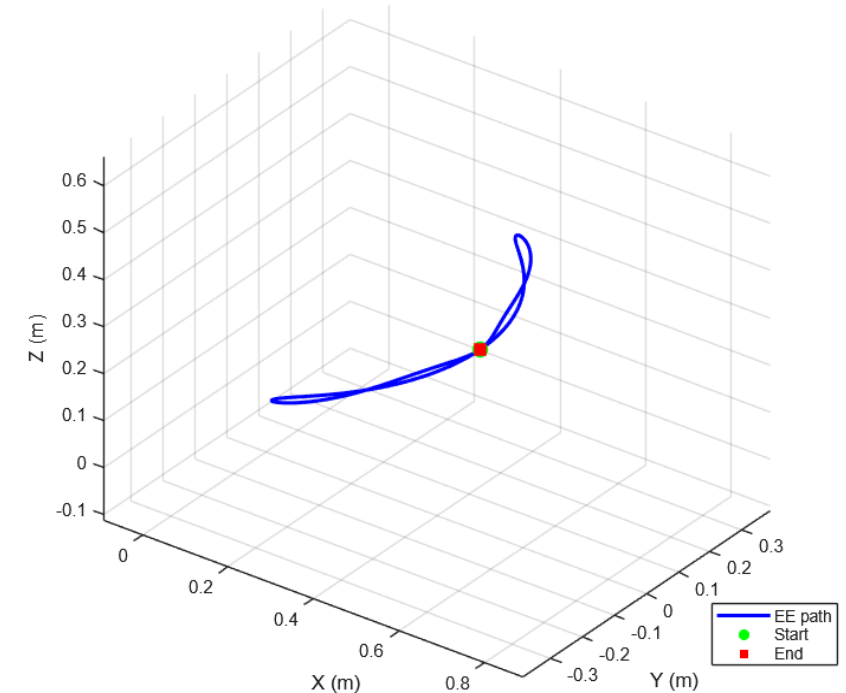
$$T_0^4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 - \theta_4) & \sin(\theta_1 + \theta_2 - \theta_4) & 0 & a_1 c_1 + a_2 c_{12} \\ \sin(\theta_1 + \theta_2 - \theta_4) & -\cos(\theta_1 + \theta_2 - \theta_4) & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$p_x = a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2)$$

$$p_y = a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2)$$

$$p_z = d_1 - d_3 - d_4$$

SCARA End-Effector Trajectory (FK simulation)



Inverse Kinematics and Reachability

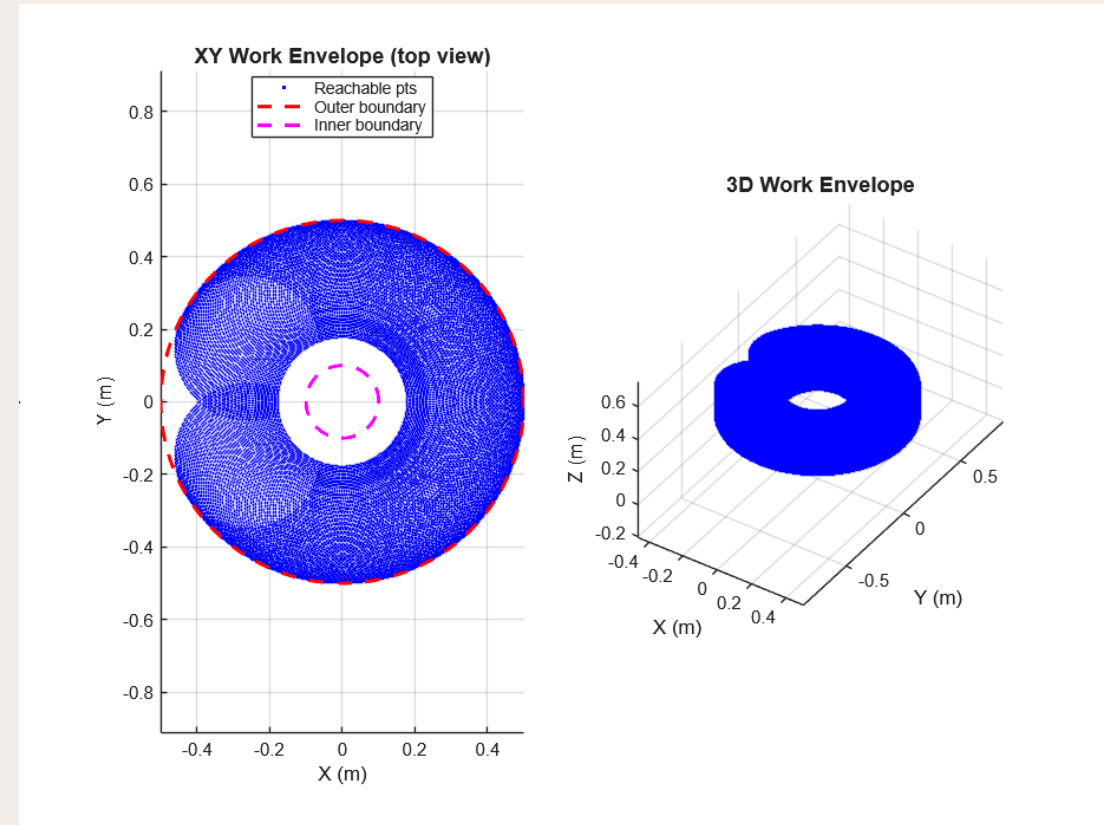
- Analytical solution used to calculate joint angles for a desired end-effector target.
- Two possible solutions for Joint 2: "Elbow-up" and "Elbow-down".
- The work envelope forms an annular ring (hollow cylinder) in 3D space.

Vertical joint (d_3): $d_3 = d_1 - p_z - d_4$

Elbow angle (θ_2): $\theta_2 = \text{atan2}(\pm\sqrt{1 - D^2}, D)$

Shoulder angle (θ_1): $\theta_1 = \text{atan2}(p_y, p_x) - \text{atan2}(a_2 \sin \theta_2, a_1 + a_2 \cos \theta_2)$

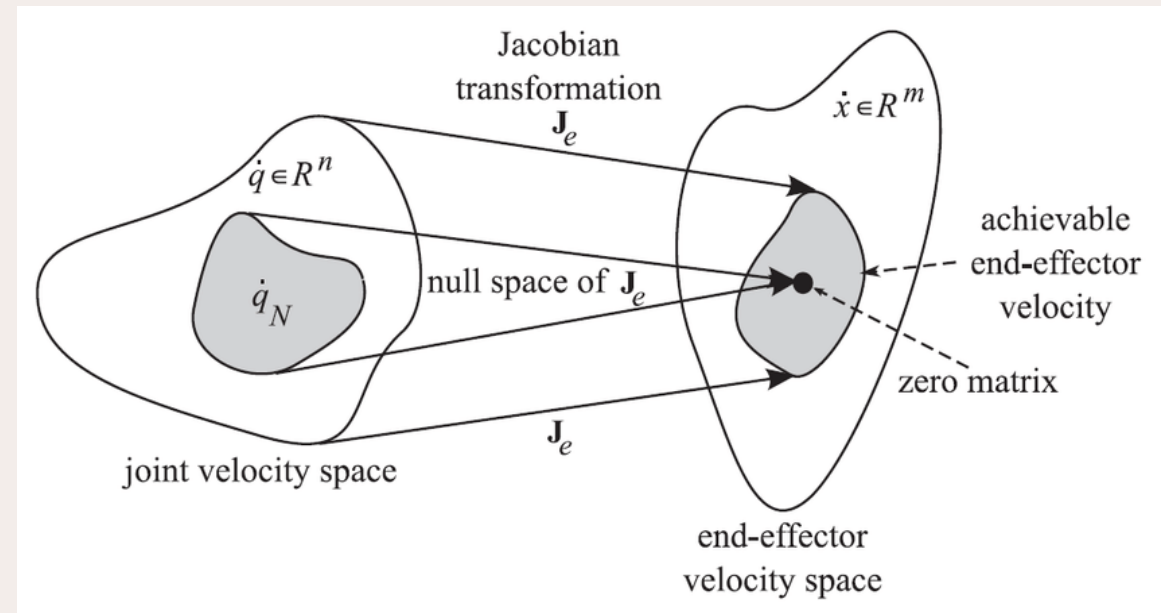
Wrist angle (θ_4): $\theta_4 = \theta_1 + \theta_2 - \phi$



Jacobian Matrix and Singularities

- 6×4 geometric Jacobian maps joint velocities to end-effector twist
- All angular velocity are only about Z-axis
- Linear velocity in XY from revolute joints 1 and 2 (meaning depends on J_{11} and J_{12}), Z -velocity is controlled by prismatic joint 3
- Singularities: Occur when $\theta_2 = 0$ or π (arm fully extended or folded), causing a loss of mobility in the radial direction.

$$\det(J_{xy}) = a_1 a_2 \sin(\theta_2)$$



Jacobian matrix mapping the joint velocity space onto the end-effector velocity space [5]

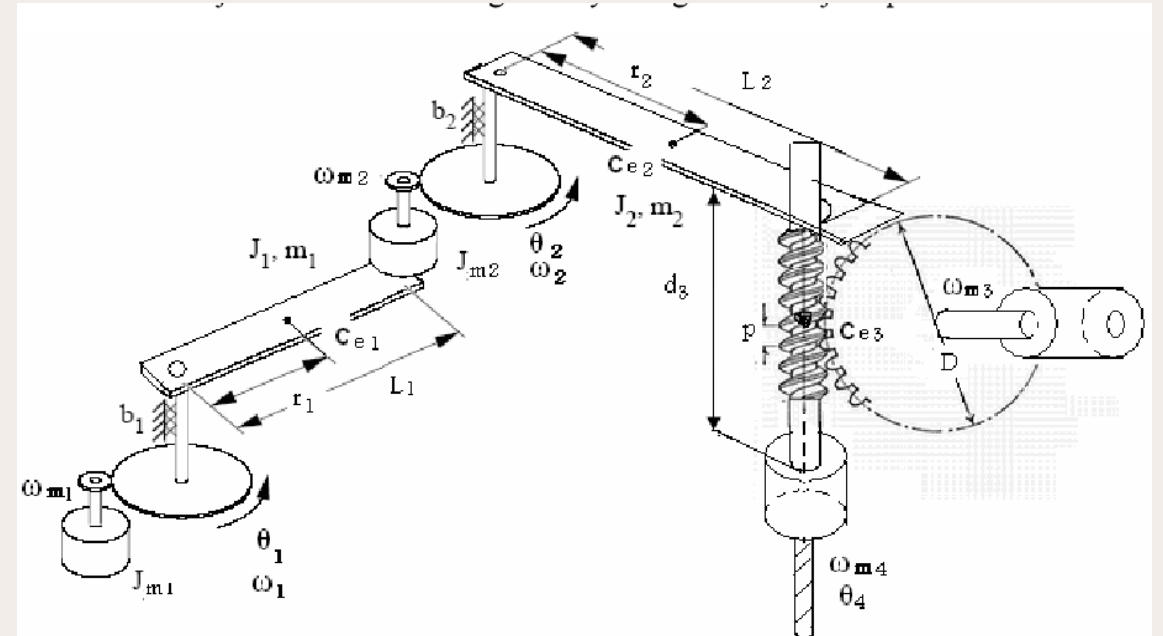
$$J = \begin{bmatrix} -(a_1 s_1 + a_2 s_{12}) & -a_2 s_{12} & 0 & 0 \\ (a_1 c_1 + a_2 c_{12}) & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$

Dynamic Modeling

- Derived using the Lagrangian ($L=T-V$) method.
- Kinetic energy (T) includes link mass and rotational inertia; Potential energy (V) is only relevant for the prismatic joint.
- Coupling: Motion in Joint 2 generates centrifugal and Coriolis forces affecting Joint 1.

Equation of motion: $\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$

$$\tau_i = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i}$$



Model of the SCARA robot: joint connections and geometry of segments and joint positions [6]

Summary of Results

- Successfully derived a complete analytical model for a 4-DOF SCARA.
- Mathematical models validated through MATLAB visualization of Forward Kinematics and work envelopes.
- Analytical inverse kinematics allow for precise trajectory planning.
- Dynamics highlight the importance of joint coupling in high-speed pick-and-place operations.

Work Cited

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- [2] "A typical SCARA structure with four joints (SCARA selective compliance assembly robot arm)," ResearchGate, fig. 1. [Online]. Available: https://www.researchgate.net/figure/A-typical-SCARA-structure-with-four-joints-SCARA-selective-compliance-assembly-robot_fig1_338407801. [Accessed: Mar. 24, 2026]. (*Typical 4-joint SCARA structure diagram*)
- [3] O. Arrayyann, "Forward-Kinematics," GitHub repository. [Online]. Available: <https://github.com/omarrayyann/Forward-Kinematics>. [Accessed: Mar. 24, 2026]. (*Forward-kinematics simulation diagram or robot chain illustration*)
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